

VIBHA BHAVIKATTI

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EDUCATION

San José State University, San Jose, California
Master of Science in Computer Science

August 2024 - June 2026 (Spring 2026)

American University of Sharjah, Sharjah, UAE
Bachelor of Science in Computer Science, Minor in Data Science

September 2020 - June 2024

Dean's List Honoree: Fall 2020, Spring 2021, Fall 2021

Chancellor's List Honoree: Fall 2021

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Distributed Systems, Machine Learning, Database Systems, Artificial Intelligence, Deep Learning, NLP, Cloud Computing, LLM, Linear Algebra, Statistics, Computer Networks

SKILLS

- **Programming:** Python, C++, Java, TypeScript, SQL, Bash
- **Backend & Infrastructure:** RESTful APIs, FastAPI, Flask, Microservices, Linux (CLI), Multithreading
- **Distributed & Data Systems:** Hadoop, Spark, Hive, HBase, ETL Pipelines, MySQL, PostgreSQL, MongoDB
- **Cloud & DevOps:** AWS (EC2, Lambda, S3), Docker, Kubernetes, CI/CD, Git
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, Reinforcement Learning, Generative AI, Computer Vision, NLP

EXPERIENCE

Scale AI

Aug 2025 - Present

Technical Advisor Intern - GenAI

- Designed **prompt variations** and agent-style evaluation tasks to analyze LLM reasoning and failure modes across edge cases and ambiguous inputs, contributing structured insights to improve generative AI reliability.
- Developed simulation-based pipelines and custom metrics using **Selenium** and **Locust** to automate evaluation of LLM outputs, enabling large-scale analysis of AI performance and integration into developer workflows.
- **Led technical reviews** of evaluation pipelines and outputs across **4 intern contributors**, identifying failure patterns and ensuring consistency in metrics, test coverage, and structured output validation.

Google Developer Groups (GDG) – SJSU

August 2025 – Present

Technical Lead

- Organize and lead AI and cloud engineering workshops for the GDG SJSU community while mentoring developers on **scalable system design** and Google Cloud technologies.

American University of Sharjah

Sept 2022 – May 2024

Teaching Assistant

- Mentored **100+ students** in Programming I, Programming Languages and Digital Systems; provided hands-on support in lab sessions and code reviews. Assisted professors with grading, documentation, debugging, and course support across 4 semesters.

PROJECTS

Neurosymbolic Compliance Engine (NEC) - Stanford TreeHacks

February 2026

Python, FastAPI, Pydantic, TypeScript, LLM Integration

- Built a **compliance engine** that evaluates regulatory rules using deterministic logic for structured constraints and LLM-based reasoning for subjective criteria.
- Modeled regulations as a knowledge graph and implemented rule parsing with dependency resolution to ensure correct evaluation order.
- Used **Pydantic** models to enforce typed schemas for inputs/outputs and validate structured LLM responses; deployed a **FastAPI backend** returning PASS/FAIL/REVIEW decisions with full audit traces.

CodeForge – Daily Coding Challenge Platform | *React, TypeScript, Firebase, Docker, AWS, CI/CD*

April 2025

- Built a full-stack **React/TypeScript web application** with **RESTful API** and **Firebase-based** user management, and session state, delivering daily coding challenges to **30+ users**.
- Developed unit + E2E tests (Vitest, Cypress) in CI/CD pipelines; enforced automated quality gates with GitHub Actions.
- Deployed using Docker containers with **AWS services** (EC2, S3) and Render, while ensuring **scalable releases and reliable service availability**.

Big Data Systems Exploration and Applied Analytics (Academic Focus)

January 2025 - May 2025

- Developed ETL pipelines and RESTful APIs in Java 11 to ingest and process multi-million record datasets across distributed systems.
- Built **indexing and retrieval workflows** using Hadoop, Spark, and Hive to optimize queries and evaluate performance trade-offs.
- Designed schemas in Hive Metastore and coordinated distributed storage (HBase, ZooKeeper) with data validation and access control for scalable, fault-tolerant access.

Traffic Sign Detection and Localization (Capstone Project) | *YOLOv3/YOLOv8, GCP, React, Python*

Oct 2023 - May 2024

- Collaborated with **iEngineering** to build a detection system for traffic signs (YOLOv3/YOLOv8), achieving mAP@0.5 of **96.1%**.
- Built a **GCP-based** platform to upload, de-duplicate, and index large annotated datasets with geo-tagged, searchable metadata.
- Optimized inference for **real-time retrieval with <1s latency** using GPU acceleration and load testing on cloud infrastructure.

COMPETITIONS

KIBO Robot Programming Competition - JAXA and NASA

April 2023 – October 2023

- Simulated and developed Astrobée's laser-based ammonia leak detection and autonomous QR code scanning; final solution was executed aboard the **International Space Station (ISS)**, achieving **1st** place in the UAE and **6th** globally.